

Project ID:

R26-IT-107

1. Topic (12 words max)

Synergizing Knowledge Graphs, Voice HCI, and Decentralized Webs for Intelligent Bookstores

2. Research group the project belongs to

Data Science, AI, & Software Engineering

3. Specialization of the project belongs to

Information Technology

4. If a continuation of a previous project:

Project ID	R26-IT-107
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature/Gravity of the Problem & User Persona: The main user personas are Sri Lankan readers, students, and upcoming authors who experience a fragmented and culturally insensitive digital literary experience. Traditional online bookstores suffer from the "Cold-Start Problem," in that new or niche Sinhala books are invisible because recommendation algorithms consider only sales history and popularity, completely disregarding the "stylistic" quality of the writing. Moreover, Language Barriers impede discovery, with most Sri Lankan users typing Sinhala titles using English letters ("Singlish") or preferring voice input, but existing search engines failing to process phonetic variations or mixed scripts and returning zero results for valid queries.

Impact of the Problem:

Technical/Social: Current platforms represent "walled gardens" that make user reviews and annotations captive within the site. This kills open academic discourse and data sovereignty.

Economic: Static pricing models penalize students by failing to anticipate academic demand, while bookstores miss out on valuable revenue due to missing the cultural demand triggers.

SDGs Alignment: The research is aligned with SDG 4 through improved access to literature facilitated by phonetic search and open annotations. It supports SDG 8 by offering the possibility of discovering emerging local authors through writing style, not only through marketing budgets.

6. Brief description of Existing Research and Systems (300 words max)

Current global giants are based or local competitors represent very high reliance on either Collaborative Filtering-as in, users who bought X also bought Y-or some very unsophisticated Content-Based Filtering, which relies on keyword matching. Search functionality involves strictly text-based search, relying on exact Unicode spelling or accurate English transliteration. Dynamic pricing in e-commerce exists but mostly reacts to inventory levels or competitor prices.

Gaps/Limitations Identified: Lack of Stylistic Intelligence: There is no existing system that models the "narrative style" or the "emotional tone" of the Sinhala literature for solving the cold-start problem.

Inflexible Search: Conventional search engines cannot handle phonetic input in "Singlish" or voice queries in mixed script, hence excluding the users not fluent in typing Sinhala Unicode.

Centralized Discourse: Reviews and reading notes are locked into the platform, which cannot be "federated" onto the open web-most recently demonstrated on Mastodon.

Blind pricing: Pricing models that abstract away from market signals, such as university semester starts, good/bad weather, or cultural events, leading to a failure in optimizing for actual demand momentum.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Let's fill the gaps with something smarter. Here's the idea: a single, intelligent bookstore platform that actually brings together four different technologies to fix what's missing.

First up, forget those tired popularity charts. SSKG-SL swaps them for a "Stylistic-Semantic Knowledge Graph," which means it recommends books based on what really matters their literary DNA and how they fit into your culture.

Next, there's PhonoLex-SL. Ever tried searching in Singlish or by voice and got nothing? PhonoLex-SL gets it. It uses a search engine that listens for how words sound, not just how they're spelled, so you can find books whether you're typing or talking.

Then you've got FedBook-Sem. Book reviews shouldn't be trapped on one site.

FedBook-Sem uses the W3C ActivityPub protocol to turn reviews into "federated" social objects. You own your data, and anyone can join the conversation, no matter where they are.

Last, MusePrice isn't just guessing at prices. It's smart. It pulls in academic calendars, weather, and other signals to predict demand and adjust prices on the fly.

So, how does this all work from start to finish?

Input:

- Raw book text for deep stylistic analysis

- User voice and Singlish text searches
- External signals, like university reading lists, weather, and social buzz

Process:

- NLP and graph tech dig into the text, pulling out stylistic features like how complex it is or how heavy it is on metaphors, then connects them to cultural info
- Phonetic tools turn Singlish or voice queries into sound-based tokens, matching them to Sinhala Unicode book titles no more getting stuck by spelling
- Federation logic transforms reviews into open web ActivityPub objects, so your thoughts travel far
- Predictive analytics models watch for signals and forecast demand, making sure prices are always up to date

Output:

- Book recommendations that actually fit your taste and context
- Search results that work, even with messy spelling
- Social features that reach across networks likes, replies, the works
- Book prices that shift with the real market, not just guesswork

Intelligent Federated Bookstore Platform

Conceptual System Architecture



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

1. Gaining Essential Specialized Domain Expertise

This project sits right where Sri Lankan Literary Criticism, Linguistics, and Decentralized Web Standards all meet. The team's about to dive deep into some pretty niche areas, and they'll pick up this expertise through focused research.

- **Literary & Cultural Context (for SSKG-SL):** The team plans to dig into Sinhala literary criticism and explore frameworks that define things like “stylistic fingerprints” think narrative rhythm or Sandhi complexity. They'll also look at cultural archetypes, like Walauwa or Gamarala, to really capture the unique feel of Sri Lankan literature in the knowledge graph.
- **Phonetics & Linguistics (for PhonoLex-SL):** For this, they're getting hands-on with Sinhala phonology and the way “Singlish” transliteration works. That means analyzing how Sri Lankans naturally use English letters to spell out Sinhala sounds, which will shape the rules for phonetic normalization.
- **Decentralized Protocols (for FedBook-Sem):** Here, the team needs to master W3C ActivityPub and Web Annotation standards. They'll do this by poring over official W3C documentation so they can set up the “Inbox/Outbox” federation system properly.

2. Gaining Technical Knowledge and Skills

The project calls for a pretty broad technical toolkit. The team's ready to pick up what they need through self-learning, online courses, and good old-fashioned technical docs.

- **Natural Language Processing (NLP) & AI:** They'll get comfortable with Sinhala NLP tools like SinLing, which they'll use to build the Stylistic Fingerprint Module. At the same time, they're diving into Graph Neural Networks to blend these stylistic features with the knowledge graph.
- **Graph Database Management:** The team's set to become fluent in Neo4j and Cypher so they can handle the complex, interconnected relationships needed for federated reviews and semantic links.
- **Machine Learning for Forecasting (for MusePrice):** They're gearing up to develop skills in time-series analysis and demand forecasting. This will help them build an engine that processes academic and environmental data to set dynamic prices.

9. Objectives and Novelty

Main Objective <i>To design and develop an intelligent online bookstore platform that integrates stylistic-semantic knowledge graphs, phonetic voice search, federated social protocols, and event-aware pricing to enhance literary discovery and accessibility in Sri Lanka.</i>			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22231482 (SSKG-SL)	<i>To design and develop a Stylistic-Semantic Knowledge Graph that integrates NLP-based style analysis with cultural ontologies to enable cold-start resistant book recommendations.</i>	1. Stylistic Fingerprint Extraction (NLP): Develop an NLP pipeline to analyze raw text for "Sandhi complexity," metaphor density, and narrative rhythm to create continuous vector embeddings of authorial voice. 2. Cultural Ontology Construction: Build a domain-specific knowledge graph mapping Sri Lankan entities (e.g., Walauwa, Gamarala) and historical eras (e.g., Post-independence).	Stylistic-Semantic Fusion: Contrary to traditional systems, which are based on sales data (CF - Collaborative Filtering) or simple keywords, "What a book feels like" is mathematically formulated, in addition to the cultural truth, making it innovative in handling the "Cold-Start Problem" for new Sinhala writers, where the ability to write is used right from Day One.

		3. Hybrid Fusion Engine: <i>Implement Graph Neural Networks (GNNs) to merge stylistic vectors with semantic graph nodes, calculating a "relevance score" based on both vibe and context.</i>	
IT22219848 (PhonoLex-SL)	To engineer a Singlish-aware phonetic and voice-driven search framework that enables accurate cross-script retrieval for Sri Lankan users.	1. Phonetic Normalization Engine: <i>Create a custom "Singlish-to-Sinhala" mapping algorithm that converts variable Latin spellings (e.g., "Amba", "Ambe") into standardized Sinhala phoneme representations.</i> 2. Native Voice Input Pipeline: <i>Implement a Speech-to-Text (STT) interface that captures spoken natural Sinhala and tokenizes it for search without requiring typing.</i> 3. Cross-Script Retrieval Algorithm: <i>Develop a search logic that matches phonetic tokens against Sinhala Unicode</i>	Acoustic-based retrieval-a paradigm of searching based on "sound similarity" as opposed to "spelling similarity" is introduced by this system. It meets a very particular challenge in HCI: Sri Lankan users struggle with Unicode keyboards or exact English transliteration. For the first time, it unifies typed Singlish and spoken Sinhala into one consistent search modality.

		<i>database titles, enabling retrieval despite script mismatches.</i>	
IT22922670 (FedBook-Sem)	<i>To design and implement a Federated Social Graph architecture that integrates social media–style interactions (reviews, comments, likes, shares) into an online bookstore using open web standards such as ActivityPub and the W3C Web Annotation Data Model, enabling decentralized, user-owned literary discourse.</i>	1. ActivityPub-Based Social Interaction Layer <i>Implement the W3C ActivityPub Inbox/Outbox architecture where both users and books function as first-class Actors. Book reviews are published as signed JSON-LD activities that behave like social media posts, allowing them to be followed, liked, replied to, and shared across federated platforms such as Mastodon, while synchronizing all interactions back to the bookstore system.</i> 2. Federated Reviews, Comments, and Likes Model <i>Replace traditional, site-locked comment sections with a federated discourse model where reviews, comments, likes, and shares are treated as interoperable social objects on the open web.</i>	<i>This federated social graph model eliminates the traditional “walled garden” approach used by commercial platforms and reimagines the online bookstore as an active participant in the decentralized social web. By combining social media style engagement with open standards, the system promotes user data sovereignty, open knowledge exchange, and interoperable academic collaboration, representing a novel application of decentralized social networking principles within e-commerce and digital humanities.</i>

		<i>This enables cross-platform engagement while preserving data ownership and preventing platform lock-in.</i> 3. Semantic Annotation System for Collaborative Reading <i>Integrate the W3C Web Annotation Data Model to support sentence-level and paragraph-level annotations within books.</i> <i>These annotations are portable, exportable, and interoperable with external scholarly tools, enabling academic discussion, textbook study, and collaborative learning beyond a single platform.</i> 4. Graph Database Implementation (Neo4j) <i>Deploy Neo4j to model complex social relationships such as follows, replies, likes, boosts, and annotation threads.</i> <i>Graph-based traversal enables efficient querying of recursive social structures (e.g., “books</i>	
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		<i>liked by people I follow across federated servers”), which is impractical in relational SQL databases.</i>	
IT22259448 (Price Change)	To develop an Adaptive Pricing Ecosystem driven by real-time academic, social, and environmental demand intelligence	1. Multi-Signal Data Ingestion: Build pipelines to ingest real-time external data, including university syllabus PDFs (academic demand), weather APIs (environmental context), and social trends. 2. Demand Forecasting Model: Train a machine learning model to predict short-term and medium-term demand spikes based on these signals, outputting a "price volatility estimate". 3. Muses Index & Smart Controls: Develop a user-facing dashboard showing price trends and a "Trade-In Futures" mechanism for students to reserve books at current prices.	Event-aware predictive pricing: In contrast to all existing dynamic pricing, which is "reactive" that is, it responds only to sales volume or stock-this system is truly "predictive." This system essentially predicts demand before it arises according to ongoing cultural signals, such as semester starts. It considers books to be "time-sensitive cultural assets" not unlike commodities on the stock market, while it balances a drive toward profitability with consumer transparency.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	
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- b) The proposed topic exhibits novelty.

Yes		No	
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- c) The student group can successfully execute the proposed project.

Yes		No	
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- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	
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- e) Any other comments:

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	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				